UNIT 3

ACCESS TO CLOUD

```
                USER
                 |
                 ▼
            CLIENT DEVICE
         Laptop / Mobile / PC
                 |
                 ▼
             WEB BROWSER
                 |
                 ▼
              INTERNET
                 |
                 ▼
       NETWORK + SECURITY
                 |
                 ▼
         CLOUD INFRASTRUCTURE
         ┌───────┼─────────┐
         ▼       ▼         ▼
      Compute  Storage   Network
         |
         ▼
      PLATFORM
         |
         ▼
   WEB APPLICATION
         |
         ▼
       WEB API
         |
         ▼
      CLOUD SERVICE
```

2. Hardware and Infrastructure

Physical infrastructure.

It includes:

Compute

- CPU
- RAM
- Servers

Storage

- HDD

- SSD
- Storage systems

Network

- Switches
- Routers
- Network Interface Cards
- Firewalls

Data-center infrastructure

- Power
- Cooling
- Physical security

A simplified diagram:

```
                CLOUD DATA CENTER
                        │
          ┌─────────────┼─────────────┐
          ▼             ▼             ▼
      COMPUTE        STORAGE       NETWORK
      Servers         SSD/HDD      Switches
      CPU/RAM         Database     Routers
```

"The hardware and infrastructure are the physical foundation on which cloud services operate."

3. Clients

A client is a device or software that requests services from a cloud system.

Examples:

- Laptop
- Desktop
- Smartphone
- Tablet
- IoT device

Example:

```
Student's Laptop
       │
       ▼
Cloud Application
```

Or:

```
Mobile Phone
     |
     ▼
Cloud Service
```

4. Security

Cloud access security includes:

- Authentication
- Authorization
- Encryption
- Firewalls
- Access control
- Identity management

Basic flow:

```
User
 |
 ▼
Username + Password
 |
 ▼
Authentication
 |
 ▼
Authorization
 |
 ▼
Cloud Service
```

For example:

```
Student
  │
  ├── View marks         ✓
  ├── View attendance    ✓
  └── Modify marks       ✗
```

5. Network

The cloud cannot be accessed without networking.

A simple example:

```
Student
   |
Laptop
   |
Wi-Fi
```

```
    |
Router
    |
Internet
    |
Cloud Data Center
```

The network provides connectivity between the client and cloud resources.

Real-time example

When a student enters:

```
www.google.com
```

the request travels through the network to Google's infrastructure.

The response then comes back to the browser.

6. Services

Cloud providers offer different types of services.

IaaS

Infrastructure as a Service

```
Virtual Machine
Storage
Network
```

PaaS

Platform as a Service

```
Runtime
Database
Development Platform
```

SaaS

Software as a Service

```
Gmail
Google Docs
Microsoft 365
```

The user doesn't necessarily manage the underlying hardware.

7. Accessing the Cloud

Suppose a student opens a cloud-based examination system.

```
Student
   │
   ▼
Laptop
   │
   ▼
Web Browser
   │
   ▼
Internet
   │
   ▼
Router / Firewall
   │
   ▼
Cloud Data Center
   │
   ▼
Web Server
   │
   ▼
Application
   │
   ▼
Database
```

This is **accessing the cloud**.

8. Platforms

A **cloud platform** provides an environment for developing, deploying, and running applications.

For example, conceptually:

```
Developer
    │
    ▼
Cloud Platform
    │
 ┌──┼─────────┐
 ▼  ▼         ▼
Runtime Database Storage
    │
    ▼
Application
```

Examples of cloud platforms include:

- Microsoft Azure
- AWS
- Google Cloud

"A platform provides the tools and environment required to build and run applications without the developer managing every physical infrastructure component."

9. Web Applications

Examples:

- Gmail
- Google Drive
- Online banking
- College ERP
- Online examination system

Architecture:

```
Browser
   │
   ▼
Web Application
   │
   ▼
Application Server
   │
   ▼
Database
```

For example, your college ERP:

```
Teacher
   │
   ▼
Browser
   │
   ▼
College ERP
   │
   ├── Student Data
   ├── Attendance
   ├── Marks
   └── Reports
```

10. Web APIs

API = Application Programming Interface

A Web API allows software applications to communicate over a network.

Example:

```
Mobile App
     │ API Request
     ▼
Cloud API
     │
     ▼
Database
     │
     ▼
API Response
     │
     ▼
Mobile App
```

11. Real-Time Example of an API

Suppose you have a weather application.

The mobile application doesn't need to maintain weather information itself.

It can request:

```
"Give me the current weather for Chennai."
```

The API sends the request to the weather service.

```
Mobile App
     │ Request
     ▼
Weather API
     │
     ▼
Weather Server
     │
     ▼
Response
     │
     ▼
Mobile App
```

The API acts as a **communication interface between applications**.

12. Web Browser

The browser is the most familiar cloud-access tool for students.

Examples:

- Google Chrome
- Microsoft Edge
- Firefox
- Safari

When you enter:

```
https://example.com
```

the browser:

1. Resolves the domain name
2. Establishes a network connection
3. Sends an HTTP/HTTPS request
4. Receives the response
5. Interprets HTML/CSS/JavaScript
6. Displays the web application

Simplified:

```
User
 │
 ▼
Browser
 │
 │ HTTPS Request
 ▼
Internet
 │
 ▼
Cloud Web Server
 │
 │ Response
 ▼
Browser
 │
 ▼
Web Page
```

13. Web Application vs Web API

This distinction is very important for exams.

Web Application	Web API
Designed primarily for users	Designed primarily for software
Usually provides UI	Usually provides programmatic interface
Browser displays the interface	Application sends/receives data

Example: Online banking website	Banking API used by mobile app
User interacts with buttons/forms	Software interacts through requests

Example:

```
              Cloud Server
                   |
       +-----------+-----------+
       |                       |
       v                       v
 Web Application           Web API
       |                       |
       v                       v
    Browser               Mobile App
```

14. Complete "Access to Cloud" Architecture

```
                      USER
                        |
                        v
                  CLIENT DEVICE
               Laptop / Mobile / PC
                        |
                        v
                   WEB BROWSER
                        |
                        v
                    INTERNET
                        |
                        v
               NETWORK + SECURITY
          +-------------+-------------+
          |                           |
          v                           v
       Firewall                    Router
          |
          v
                CLOUD DATA CENTER
       +---------------+---------------+
       |               |               |
       v               v               v
    Compute         Storage         Network
       |
       v
   CLOUD PLATFORM
       |
       v
 +-----+-----------+
 |                 |
 v                 v
Web Application   Web API
 |                 |
 v                 v
Browser           Mobile App
 |                 |
 +--------+--------+
          |
          v
     CLOUD SERVICES
```

15. One Complete Real-Time Example

Use a **Google Drive-like application** to explain the entire topic.

A student wants to upload an assignment.

Step 1 – Client

```
Student
   ↓
Laptop
```

Step 2 – Browser

```
Laptop
   ↓
Chrome / Edge
```

Step 3 – Network

```
Browser
   ↓
Wi-Fi
   ↓
Router
   ↓
Internet
```

Step 4 – Security

```
HTTPS
   ↓
Authentication
   ↓
Authorization
```

Step 5 – Cloud

```
Cloud Data Center
   ↓
Web Application
   ↓
Storage
```

Step 6 – Upload

```
Assignment
     ↓
Web Application
     ↓
Web API
     ↓
Cloud Storage
```

Step 7 – Response

```
Cloud Storage
     ↓
Web API
     ↓
Web Application
     ↓
Browser
     ↓
"Upload Successful"
```

MCQs – BL3, BL4 & BL5

Topics: Hardware & Infrastructure, Clients, Security, Network, Services, Accessing Cloud, Platforms, Web Applications, Web APIs, Web Browsers

BL3 – Apply

Q1. [BL3]

A university wants to host an online examination system in the cloud. The system requires CPU, RAM, storage, and networking resources. Which infrastructure component primarily provides the processing capability?

A. Storage array
B. Compute server
C. Firewall
D. DNS server

Answer: B. Compute server

Explanation: Compute servers provide processing resources such as **CPU and RAM** required to run cloud applications and virtual machines.

Q2. [BL3]

A student accesses a cloud-based learning management system using a laptop connected through Wi-Fi. In this scenario, the laptop primarily acts as a:

A. Cloud provider
B. Client
C. Hypervisor
D. Storage server

Answer: B. Client

Explanation: A **client** is a device or software that requests and consumes services provided by a cloud system.

Q3. [BL3]

A user enters a username and password before accessing a cloud application. Which security mechanism is being applied?

A. Authorization
B. Authentication
C. Load balancing
D. Data replication

Answer: B. Authentication

Explanation: Authentication verifies **who the user is**. Authorization determines what that authenticated user is allowed to access.

Q4. [BL3]

A student accesses a cloud application from Chennai using a laptop connected to the Internet. Which component provides connectivity between the laptop and the cloud data center?

A. Network
B. Database
C. CPU
D. Application server

Answer: A. Network

Explanation: The network provides communication between the user's client device and cloud infrastructure through technologies such as Wi-Fi, routers, Internet protocols, and network links.

Q5. [BL3]

A company wants to use a virtual machine, virtual storage, and virtual networking without purchasing physical servers. Which cloud service model should it select?

A. SaaS
B. PaaS

C. IaaS
D. Web API

Answer: C. IaaS

Explanation: Infrastructure as a Service (IaaS) provides virtualized compute, storage, and networking resources.

Q6. [BL3]

A user opens Microsoft Edge and enters the URL of a cloud-based application. Which software is directly used to display the web application?

A. Hypervisor
B. Web browser
C. Database server
D. Storage controller

Answer: B. Web browser

Explanation: A web browser such as Edge, Chrome, or Firefox sends requests to web servers and displays the returned web content.

Q7. [BL3]

A mobile application needs to retrieve a student's attendance information from a cloud server. Which component should it use to communicate programmatically with the cloud application?

A. Web API
B. Web browser only
C. Physical switch
D. RAM

Answer: A. Web API

Explanation: A **Web API** provides a programmatic interface through which applications can communicate with cloud services.

BL4 – Analyze

Q8. [BL4]

Consider the following cloud access path:

```
Laptop → Wi-Fi Router → Internet → Cloud Data Center → Web Server
```

A user can connect to the Internet but cannot access the cloud application. Which component should be investigated first if other websites are also inaccessible?

A. Cloud database
B. Network connectivity
C. Web application code
D. Cloud storage

Answer: B. Network connectivity

Explanation: If multiple websites are inaccessible, the problem is likely related to network connectivity rather than the specific cloud application.

Q9. [BL4]

A user successfully logs into a cloud application but receives a message saying:

"You are not permitted to view this resource."

Which process is responsible for this decision?

A. Authentication
B. Authorization
C. DNS resolution
D. Encryption

Answer: B. Authorization

Explanation: Authentication establishes the user's identity. **Authorization determines whether that user has permission to access a particular resource.**

Q10. [BL4]

A cloud application consists of:

```
Browser
   ↓
Web Server
   ↓
Application Server
   ↓
Database
```

The browser displays the application's interface, while the application server performs business logic. Which component primarily acts as the user-facing interface?

A. Database
B. Application server
C. Web browser
D. Storage system

Answer: C. Web browser

Explanation: The browser provides the user interface through which users interact with the web application.

Q11. [BL4]

A cloud provider notices that thousands of users are accessing an application simultaneously. The application becomes slow because a single server is processing all requests.

Which approach would best address the problem?

A. Increase the monitor resolution
B. Use load balancing and additional compute resources
C. Remove network security
D. Reduce storage capacity

Answer: B. Use load balancing and additional compute resources

Explanation: Load balancing distributes requests across multiple servers, while additional compute resources increase processing capacity.

Q12. [BL4]

A developer wants to build a cloud application without managing physical servers or operating-system installation. The developer wants access to runtime, databases, and development tools.

Which cloud service model best matches the requirement?

A. IaaS
B. PaaS
C. SaaS
D. LAN

Answer: B. PaaS

Explanation: **PaaS** provides a managed development and deployment environment, including runtime and often databases and development tools.

Q13. [BL4]

A cloud-based banking application uses HTTPS when users submit login credentials. What is the primary security benefit?

A. It increases CPU capacity
B. It protects communication between the client and server
C. It increases storage capacity
D. It eliminates the need for authentication

Answer: B. It protects communication between the client and server

Explanation: HTTPS uses encryption to protect data exchanged between the browser and server from interception or tampering.

Q14. [BL4]

A mobile application retrieves student information using:

```
Mobile App → HTTP Request → Cloud API → Database
```

The API returns the requested information to the mobile application.

What is the primary role of the API?

A. Store the mobile application's screen
B. Provide an interface for communication between software components
C. Replace the physical network
D. Provide CPU hardware

Answer: B. Provide an interface for communication between software components

Explanation: A Web API acts as a communication interface, allowing applications to request and exchange data with cloud services.

BL5 – Evaluate

Q15. [BL5]

A hospital wants to deploy a mission-critical cloud application. It requires high availability, secure networking, scalable compute resources, and reliable storage.

Which architecture is the most appropriate?

A. Single desktop computer with local storage
B. Cloud infrastructure with redundant compute, storage, networking, and security controls
C. One physical server without backup
D. Standalone application without networking

Answer: B. Cloud infrastructure with redundant compute, storage, networking, and security controls

Explanation: Mission-critical applications require **redundancy, scalability, security, and reliable infrastructure**. A properly designed cloud infrastructure provides these capabilities.

Q16. [BL5]

A university wants students to access an online learning platform from laptops, tablets, and smartphones. The application should not require users to install special software.

Which approach is most suitable?

A. Browser-based web application
B. Physical server console
C. Desktop-only application
D. Local USB application

Answer: A. Browser-based web application

Explanation: A browser-based cloud application can be accessed from multiple client devices without requiring specialized application installation.

Q17. [BL5]

A company is selecting between IaaS and PaaS for developing a new web application. Its developers want to focus on application code and do not want to manage operating systems or runtime environments.

Which option should they select?

A. IaaS
B. PaaS
C. Physical hosting
D. Local storage

Answer: B. PaaS

Explanation: PaaS abstracts infrastructure and platform management, allowing developers to concentrate primarily on application development.

Q18. [BL5]

An organization wants to protect a cloud application against unauthorized access. Which combination provides the strongest overall approach?

A. Password only
B. Authentication, authorization, encryption, and network security
C. Increasing CPU capacity
D. Increasing storage capacity

Answer: B. Authentication, authorization, encryption, and network security

Explanation: Cloud security requires **multiple layers**. Authentication verifies users, authorization controls permissions, encryption protects data, and network security protects communication and infrastructure.

Q19. [BL5]

A company wants its mobile app, web application, and third-party applications to access the same cloud-based customer service. Which architecture would be most appropriate?

A. Separate databases with no communication
B. Cloud backend exposed through well-designed Web APIs
C. One desktop application installed on every device
D. Direct access to the cloud database by every client

Answer: B. Cloud backend exposed through well-designed Web APIs

Explanation: APIs provide a controlled and reusable interface through which different applications can access backend cloud services without directly accessing the database.

Q20. [BL5]

An organization wants to provide a cloud-based examination system for 50,000 students. The system must support browser access, secure communication, scalable infrastructure, and communication between frontend and backend services.

Which design is the BEST choice?

A. One physical server with a desktop application
B. Browser → HTTPS → Load Balancer → Scalable Web/Application Servers → Web API → Database/Storage
C. Student computers directly accessing the database
D. Browser → USB storage → Application

Answer: B. Browser → HTTPS → Load Balancer → Scalable Web/Application Servers → Web API → Database/Storage

Explanation: This architecture addresses all requirements:

SECOND HALF OF UNIT 3

1. Cloud Storage – Overview

Basic architecture

```
                  USER
                        |
                        ▼
                  Cloud Application
                        |
                        ▼
            Storage Service
                        |
            ┌───────┼───────┐
            ▼     ▼       ▼
            Disk     SSD       Storage Pool
```

Cloud storage provides storage capacity **on demand**, without the user having to manage the underlying physical disks.

Types of Cloud Storage

1. Object Storage

Best for:

- Images
- Videos
- Documents
- Backups
- Large unstructured data

```
Application
      |
      ▼
Object Storage
 ┌───┼────┐
```

```
 ▼   ▼      ▼
File File File
```

Example conceptually: storing thousands of student assignment PDFs.

2. Block Storage

```
VM
 |
 ▼
Virtual Disk
 |
 ▼
Block Storage
```

It is commonly used for:

- Operating systems
- Databases
- Application disks

3. File Storage

Provides a shared file system.

```
VM1 ──┐
VM2 ──┼──► Shared File Storage
VM3 ──┘
```

Multiple systems can access shared files according to permissions.

2. Storage Area Network (SAN) Architecture

SAN is a high-speed specialized network that connects servers to shared storage systems.

Architecture

```
            SERVERS
      ┌───────┼──────┐
      ▼       ▼    ▼
```

```
Server Server Server
|       |     |
└───────┼─────┘
       |
SAN SWITCH
       |
┌───────┼─────┐
▼       ▼     ▼
Storage Storage Storage
Array    Array    Array
```

SAN separates storage traffic from the normal user/application network.

Real-Time Example

Suppose a cloud data center has:

```
100 Physical Servers
10 TB Storage
```

Instead of giving each server isolated storage:

```
Server 1 → Disk 1
Server 2 → Disk 2
Server 3 → Disk 3
```

SAN allows:

```
Server 1 ──┐
Server 2 ──┤
Server 3 ──┼──▶ SAN ──▶ Storage Array
Server 4 ──┤
Server 5 ──┘
```

This makes centralized storage management possible.

3. Collaboration Within the Cloud

```
Student A ──┐
Student B ──┼──▶ Cloud Document
Student C ──┘
```

This is **cloud collaboration**.

Users can:

- Share files
- Edit documents
- Comment
- Track changes
- Work simultaneously
- Control permissions
  Example

- Google Docs
- Microsoft 365
- Online project management tools

How Collaboration Works

```
User A ─┐
User B ─┼──► Cloud Application
User C ─┘          |
                   ▼
              Cloud Storage
                   |
                   ▼
            Shared Document
```

The cloud provides a **central location** where multiple authorized users can work with the same data.

4. Standards

A standard defines common rules for communication, data exchange, security, or service interaction.

Examples of technologies/protocols students may encounter:

- HTTP / HTTPS
- REST
- SOAP
- XML
- JSON
- TCP/IP
- OAuth
- SAML

Standards allow different systems and applications to communicate in a predictable and interoperable way.

5. Driving Forces of Cloud Computing

1. Cost Reduction

Instead of buying large amounts of hardware:

```
Traditional:

Buy Server
```

```
Buy Storage
Buy Network
Maintain Everything
```

Cloud:

```
Use Resources
      ↓
Pay for What You Use
```

2. Scalability

Suppose an online exam normally has:

```
1,000 users
```

But during the exam:

```
10,000 users
```

Cloud resources can be increased.

```
Before:
VM1 + VM2

High demand:
VM1 + VM2 + VM3 + VM4 + VM5
```

3. Availability

Cloud providers can use multiple servers/data centers to improve availability.

```
Server A ──┐
Server B ──┼──► Cloud Service
Server C ──┘
```

If one server fails, another can continue providing service.

4. Flexibility

Resources can be created, resized, or removed quickly.

5. Global Access

Users can access cloud services from different locations through the Internet.

6. Rapid Deployment

Applications can be deployed much faster than traditional hardware-based infrastructure.

6. Software and Services

Cloud provides different layers of services.

```
            SaaS
      Applications for Users
            |
            ▼
            PaaS
      Development Platform
            |
            ▼
            IaaS
      Compute + Storage + Network
            |
            ▼
      Physical Infrastructure
```

IaaS

User gets infrastructure.

Example:

```
VM
Storage
Network
```

PaaS

Developer gets a platform.

```
Runtime
Database
Development Tools
```

SaaS

End user gets a complete application.

```
Email
Office Suite
CRM
ERP
```

7. Developing Applications in the Cloud

"How can a developer create an application that runs in the cloud?"

Basic process:

```
Developer
      |
      ▼
```

```
Cloud Platform
      |
      ▼
Write Application
      |
      ▼
Use APIs
      |
      ▼
Connect Database
      |
      ▼
Deploy Application
      |
      ▼
Cloud Users
```

Example: Cloud-Based Student Management System

Suppose students want to develop:

Cloud-Based Student Attendance System

Front end

```
Web Browser
      |
      ▼
Web Application
```

Backend

```
Application Server
      |
      ▼
Web API
```

Database

```
Database
      |
      ▼
Cloud Storage
```

Complete architecture:

```
                  STUDENT
                        |
                        ▼
                  Browser
                        |
```

```
                  ▼
      Web Application
                  |
                  ▼
            Web API
                  |
                  ▼
      Application Server
                  |
                  ▼
            Database
                  |
                  ▼
            Cloud Storage
```

9. Real-Time Example: Google Drive

Suppose you upload a project report to Google Drive.

Cloud Storage

Your file is stored in cloud infrastructure.

Collaboration

You share it with your classmates.

```
You ──┐
      ├──► Shared Document
Friend ┘
```

Standards

Your browser communicates using Internet/web protocols.

Services

You are consuming a cloud-based software service.

Application Development

The underlying cloud platform uses software, APIs, databases, storage, and networking to provide the service.

11. Very Important Distinction: SAN vs Cloud Storage

Students often confuse these.

SAN

Infrastructure technology

```
Servers
   |
   ▼
SAN
   |
   ▼
Storage Array
```

It provides high-speed access to centralized storage.

Cloud Storage

Cloud service

```
User
  |
  ▼
Cloud Storage Service
  |
  ▼
Underlying Storage Infrastructure
```

The cloud provider manages the underlying infrastructure.

"SAN is one technology that can be used inside the infrastructure supporting cloud storage; cloud storage is the service exposed to users."

BL3 – APPLY

Q1. Cloud Storage

A university wants to store thousands of student project videos. The files are large, unstructured, and need to be accessed through a cloud application. Which storage approach is most suitable?

A. Block storage
B. Object storage
C. CPU cache
D. Register storage

Answer: B. Object storage

Explanation: Object storage is well suited for large amounts of unstructured data such as videos, images, documents, and backups.

Q2. Block Storage

A cloud administrator creates a virtual machine and wants to attach a virtual disk to install an operating system and database. Which storage type is most appropriate?

A. Object storage
B. Block storage
C. Email storage
D. Archive tape only

Answer: B. Block storage

Explanation: Block storage provides virtual disk-like storage and is commonly used for VM operating systems, databases, and applications requiring low-level disk access.

Q3. SAN Architecture

A cloud data center has 200 physical servers that need high-speed access to a centralized storage system. Which architecture is most appropriate?

A. SAN
B. PAN
C. Desktop storage
D. Local USB storage

Answer: A. SAN

Explanation: A **Storage Area Network (SAN)** provides high-speed network-based access between servers and centralized storage arrays.

Q4. Cloud Collaboration

Three students are simultaneously editing the same project document stored in the cloud. Which cloud capability is being demonstrated?

A. Virtualization
B. Cloud collaboration
C. Hypervisor isolation
D. CPU scheduling

Answer: B. Cloud collaboration

Explanation: Cloud collaboration allows multiple authorized users to access, edit, comment on, and share common cloud-hosted resources.

Q5. Cloud Standards

A developer wants a mobile application and cloud server to exchange structured data in a commonly understood format. Which technology is commonly suitable?

A. JSON
B. CPU instruction set
C. RAM
D. SATA cable

Answer: A. JSON

Explanation: JSON is a widely used data-interchange format for communication between applications and web/cloud services.

Q6. Driving Forces

An online examination normally has 1,000 users but occasionally increases to 20,000 users. Which cloud characteristic is most important for handling this variation?

A. Scalability
B. Physical isolation
C. Manual configuration
D. Fixed hardware capacity

Answer: A. Scalability

Explanation: Cloud scalability allows computing resources to increase or decrease according to workload demand.

Q7. Cloud Services

A developer wants an environment containing runtime software, databases, and development tools without managing physical servers. Which service model is most appropriate?

A. IaaS
B. PaaS
C. SaaS
D. LAN

Answer: B. PaaS

Explanation: Platform as a Service (PaaS) provides an application development and deployment environment while hiding much of the underlying infrastructure management.

BL4 – ANALYZE

Q8. SAN vs Local Storage

A company currently gives every application server its own local storage. Storage utilization is poor, and replacing failed disks requires manual intervention. The company wants centralized storage that can be accessed by multiple servers.

Which change best addresses the problem?

A. Replace all servers with desktop computers
B. Implement a SAN with centralized storage arrays
C. Remove all storage from servers
D. Use only CPU virtualization

Answer: B. Implement a SAN with centralized storage arrays

Explanation: A SAN provides centralized storage that can be accessed by multiple servers. It improves storage management and resource utilization compared with isolated local disks.

Q9. Storage Performance

A cloud application uses a database that performs frequent read/write operations. The application experiences high latency when using object storage.

Which storage type would generally be more appropriate?

A. Block storage
B. Object storage only
C. Web browser cache
D. Email storage

Answer: A. Block storage

Explanation: Block storage is designed for applications such as databases and VM disks that require low-level, frequent read/write operations.

Q10. Cloud Collaboration

A research team stores a document locally on each member's laptop. Every time someone makes a change, all members must manually exchange updated copies.

Which cloud-based solution best solves the problem?

A. Centralized shared cloud storage with collaboration features
B. Increasing CPU speed

C. Installing a hypervisor
D. Creating separate physical servers for each user

Answer: A. Centralized shared cloud storage with collaboration features

Explanation: Cloud collaboration provides a common, centrally managed copy that authorized users can access and modify, reducing version conflicts and manual file exchange.

Q11. Standards and Interoperability

A company develops an application that needs to communicate with services from multiple cloud providers. Each provider uses different proprietary interfaces, making integration difficult.

Which concept would most directly help improve interoperability?

A. Common standards and standardized APIs
B. Increasing RAM
C. Increasing disk capacity
D. Using a larger monitor

Answer: A. Common standards and standardized APIs

Explanation: Standards and standardized APIs provide common communication mechanisms, making it easier for applications and services from different providers to interoperate.

Q12. Cloud Driving Forces

A university hosts an admission portal. During normal periods it needs 5 servers, but during admission deadlines it requires 50 servers. After the deadline, demand falls back to 5 servers.

Which combination of cloud characteristics best addresses this situation?

A. Scalability and elasticity
B. Physical isolation and SAN
C. Manual provisioning and fixed capacity
D. Local storage and desktop computing

Answer: A. Scalability and elasticity

Explanation: Scalability allows capacity to increase, while **elasticity** allows resources to dynamically expand and contract according to demand.

Q13. IaaS vs PaaS

A development team wants to deploy a Python web application. They do not want to manage operating system patches, runtime installation, or physical servers.

Which approach is more suitable?

A. IaaS
B. PaaS
C. Bare-metal hosting
D. Local desktop computing

Answer: B. PaaS

Explanation: PaaS provides a managed application development and deployment environment, reducing the developer's responsibility for infrastructure and platform maintenance.

Q14. Cloud Application Architecture

A student develops a cloud-based attendance application using:

- Browser-based frontend
- Web API
- Application server
- Cloud database

Which component should primarily act as the communication interface between the frontend and backend services?

A. Web API
B. SAN switch
C. CPU cache
D. Physical hard disk

Answer: A. Web API

Explanation: A **Web API** provides an interface through which the frontend can communicate with backend application services.

BL5 – EVALUATE

Q15. Choosing Storage

A media company needs to store millions of images and videos. The data does not require a traditional file-system structure, and the company wants highly scalable cloud storage.

Which solution should be preferred?

A. Object storage
B. CPU registers
C. Local RAM
D. Block storage for every individual file

Answer: A. Object storage

Explanation: Object storage is highly suitable for massive collections of unstructured data such as images and videos and can scale to very large capacities.

Q16. Choosing SAN

A cloud data center runs hundreds of database servers. The databases require high-performance shared storage, centralized management, and reliable storage connectivity.

Which solution is most appropriate?

A. Individual USB disks
B. SAN with centralized storage arrays
C. Laptop hard drives
D. Browser cache

Answer: B. SAN with centralized storage arrays

Explanation: SAN is designed for high-speed connectivity between servers and centralized storage systems, making it suitable for demanding enterprise workloads.

Q17. Evaluating Cloud Collaboration

A multinational company has teams working from India, Japan, and Germany. They need to work on the same project documents without maintaining multiple copies.

Which solution provides the greatest benefit?

A. Local storage on every employee's computer
B. Cloud-based shared storage and collaboration tools
C. Separate physical servers without synchronization
D. Offline file exchange using USB drives

Answer: B. Cloud-based shared storage and collaboration tools

Explanation: Cloud collaboration allows geographically distributed teams to access common resources and work together in near real time.

Q18. Selecting a Cloud Service Model

A startup wants to develop and deploy a web application quickly. It does not want to purchase hardware or manage operating systems but needs control over its application code and data.

Which cloud service model is the best choice?

A. SaaS
B. PaaS
C. Traditional physical hosting
D. Local storage

Answer: B. PaaS

Explanation: PaaS provides the development and deployment environment while reducing infrastructure and operating-system management responsibilities.

Q19. Cloud Application Development

A company wants to develop a scalable online shopping application. It requires a web interface, backend services, database, cloud storage, and APIs.

Which architecture is the most appropriate?

A. Single standalone desktop application
B. Cloud-based multi-tier application using APIs and managed cloud services
C. Application running only from a USB drive
D. One physical computer without networking

Answer: B. Cloud-based multi-tier application using APIs and managed cloud services

Explanation: A multi-tier cloud architecture separates the presentation, application, API, database, and storage layers, making the system easier to scale and maintain.

Q20. Overall Cloud Evaluation

A university wants to build a cloud-based learning platform with:

- Thousands of students
- Shared course materials
- Video lectures
- Online assignments
- Collaboration between teachers and students
- Variable workload during examinations

Which combination provides the most suitable solution?

A. Local PCs + individual storage + manual file sharing
B. Cloud storage + collaboration services + scalable infrastructure + web APIs

C. One physical server + fixed storage + no networking
D. Individual USB drives + desktop applications

Answer: B. Cloud storage + collaboration services + scalable infrastructure + web APIs

Explanation: The proposed solution addresses all requirements: cloud storage handles course materials, collaboration supports teachers and students, scalability handles varying workloads, and web APIs allow different application components to communicate.